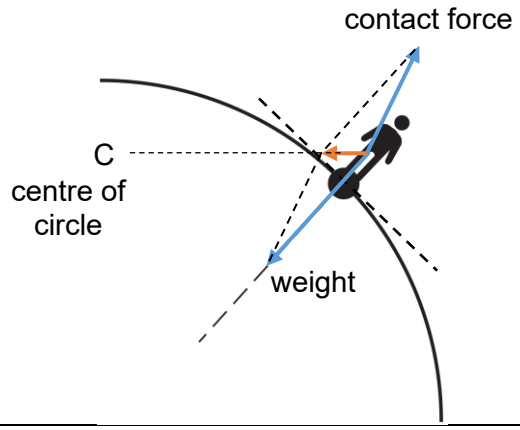
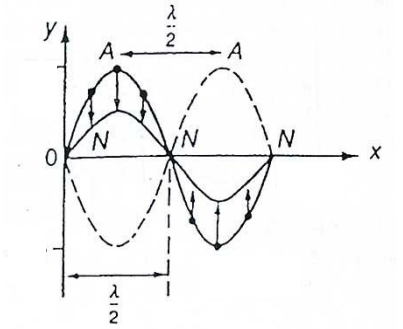
Anderson Serangoon Junior College 2026 H2 Physics Preliminary Exam Scheme

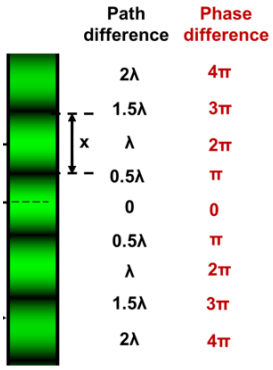
Paper 2 (75 marks)

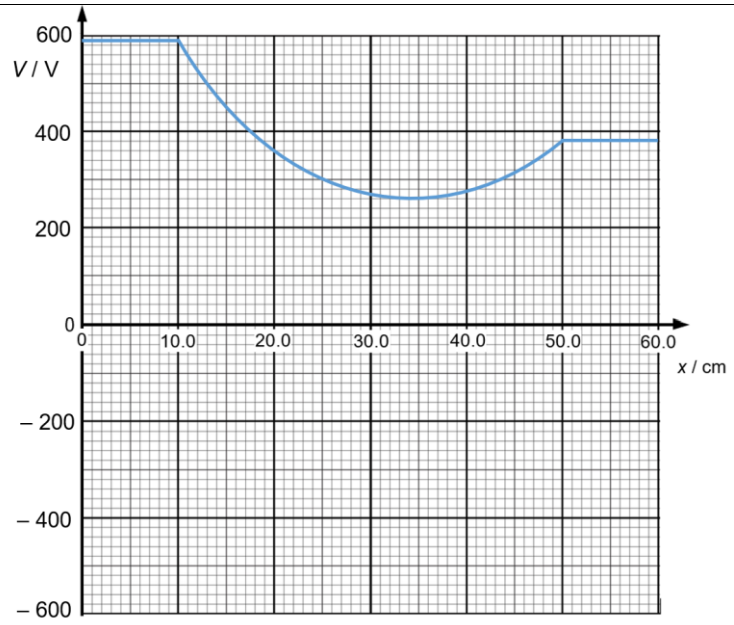
1ai	Resultant force = centripetal force = $mr\omega^2 = m(R\cos 52.2)\left(\frac{2\pi}{T}\right)^2$ = $58.6\times 6.37\times 10^6 \cos 52.2\times\left(\frac{2\pi}{24\times 60\times 60}\right)^2$ = 1.21 N	C1 C1 A1
1aii	arrow from student pointing along the dotted line, labelled 'weight' upwards arrow from student pointing in a direction to the left of normal and above the tangent to the Earth, labelled 'contact force'  	B1 B1
1b	Apparent weight (which is the normal contact force) is the difference between gravitational force and the component of centripetal force along the radial direction. For same angular velocity, larger radius means larger centripetal force required at equator. Hence, apparent weight is smaller at equator.	M1 A1
2ai	Gravitational force provides centripetal force $\frac{GMm}{r^2}=\frac{mv^2}{r}$ and $v=\frac{2\pi r}{T}$ OR $\frac{GMm}{r^2}=mr\omega^2$ and $\omega=\frac{2\pi}{T}$ $r=\sqrt[3]{\frac{6.67\times 10^{-11}\times 6.0\times 10^{24}\times(13.7\times 24\times 3600)^2}{4\pi^2}}=2.4217\times 10^8$ m $r\approx 2.4\times 10^8$ m	B1 C1 A1 A0

2aii	Change in GPE $= -\frac{GMm}{r_{orbit}} - \left(-\frac{GMm}{r_{surface}}\right) = -GMm\left(\frac{1}{r_{orbit}} - \frac{1}{r_{surface}}\right)$ $= -6.67\times10^{-11}\times6.0\times10^{24}\times360\times\left(\frac{1}{2.4\times10^{8}} - \frac{1}{6.4\times10^{6}}\right)$ $= 2.2 \times 10^{10}$ J	C1 A1
2b	(Rock has minimum speed when all its KE is converted to GPE to reach infinity with zero GPE and zero KE) Loss in KE of rock = Gain in GPE or $\frac{1}{2}mv^2 = \frac{GM_{moon}m}{R_{moon}}$ $v_{\min} = \sqrt{\frac{2GM_{moon}}{R_{moon}}} = \sqrt{\frac{2\times6.67\times10^{-11}\times7.4\times10^{22}}{1.7\times10^{3}\times10^{3}}} = 2.4\times10^{3}\ \text{m s}^{-1}$ Since $V_{rock} > V_{\min}$, it will travel out into distant space **<u>Alternative methods</u>** Change in GPE from Moon surface to infinity $= \frac{GM_{moon}m}{R_{moon}} = \frac{6.67\times10^{-11}\times7.4\times10^{22}}{1.7\times10^{3}\times10^{3}}m = 2.9\times10^{6}\ m$ KE initial = ½ mv^2 = ½ x m x $(2.8 \times 10^3)^2$ = 3.9×10^6 m Since KE initial > change in GPE from surface to infinity, it will travel out into distant space.	M1 M1 A1 (M1) (M1) (A1)
3a	the *<u>increase</u>* <u>in internal energy</u> of a system is equal to the <u>sum</u> of the *<u>heat supplied to</u>* the system and the *<u>work done</u>* *<u>on</u>* the system. Only get 1 mark if candidate did not use the word “system”.	A1
3b	At <u>constant volume</u>, <u>no work is done</u> by the gas. <u>Heat</u> required <u>is solely used to</u> increase <u>its internal energy</u>. At constant pressure, work is done by the gas as it expands. <u>Heat is required for the increase of its internal energy and the work done by the gas</u>. More heat is required to increase the temperature of unit mass of gas by one unit for constant pressure. From Q = mcΔθ, more heat implies larger specific heat capacity.	B1 B1 A0
3c	Internal energy consists of potential energy due to intermolecular forces and kinetic energy due to random motion. Temperature is only making reference to the kinetic energy due to random motion. Temperature is proportional to the average kinetic energy due to random motion but internal energy also depends of the total number of particles in the system.	B1 B1

3di1	Equation of state: $pV = nRT$ $2.62\times10^5(0.0360) = n(8.31)(273.15+25)$ $n = 3.8069 \text{ mol} = 3.8 \text{ mol}$	M1
3di2	$pV = nRT$ $p(0.0360) = (0.30+3.8069)(8.31)(273.15+25)$ $p = 2.8217\times10^5 \text{ Pa}$	C1 A1

4a	It is a wave in which the oscillations of the particles in the wave are along the direction of transfer of energy of the wave.	B1
4bi	phase difference = 180° or antiphase  	B1
4bii	λ/2 = 25.0 cm or 0.250 m $v = f\lambda$ $f = v/\lambda$ = 330/0.500 = 660 Hz	C1 A1
4biii	A^2 readings from the graph: 2.6 and 4.0 Since intensity α A^2, ratio = 2.6/4.0 = 0.65	C1 A1
5a	When two (or more) waves meet at a point, the resultant displacement is equal to the vector sum of the individual displacements.	M1 A1
5bi	The sources are in-phase at the double-slit (Since path difference (from slits to A) is zero) the phase difference is zero / the waves meet in-phase so constructive interference at A.	B1 B1
5bii	$x = \frac{\lambda D}{a}$ $= \frac{(546\times10^{-9})(0.85)}{0.13\times10^{-3}} = 3.57\times10^{-3}\text{ m}$ AB = 2.5x = 2.5 (3.57 x 10^{-3})	C1 C1

	= 8.9 x 10^{-3} m  	A1
5biii	The wavelength for blue light is shorter, so the separation is less.	B1
5biv	Maxima/ bright fringe(s)/ central bright fringe/ fringe at A is less bright Minima/ dark fringe(s) is brighter Contrast between maxima and minima (or fringes) decreases	B2
6a	field strength (at a point) is equal to potential gradient / gradient of potential-distance graph (at that point) negative sign included or discussed **/** field strength acts in the direction of decreasing potential	B1 B1
6bi	horizontal solid lines, and non-zero potential within the spheres positive concave curve between A and B minimum nearer to sphere B, i.e. x > 30.0 cm values of V_{net} at 10.0 cm and 50.0 cm calculated and plotted with correct unit value of net potential on the surface of A $= \frac{Q_A}{4\pi\varepsilon_o r_A} + \frac{Q_B}{4\pi\varepsilon_o r_B} = \frac{1}{4\pi \times 8.85\times10^{-12}}\left(\frac{6.0\times10^{-9}}{10\times10^{-2}} + \frac{3.0\times10^{-9}}{50\times10^{-2}}\right) = 593.45 \approx 590\ \text{V}$ value of net potential on the surface of B $= \frac{Q_A}{4\pi\varepsilon_o r_A} + \frac{Q_B}{4\pi\varepsilon_o r_B} = \frac{1}{4\pi \times 8.85\times10^{-12}}\left(\frac{6.0\times10^{-9}}{50\times10^{-2}} + \frac{3.0\times10^{-9}}{10\times10^{-2}}\right) = 377.66 \approx 380\ \text{V}$	B1 B1 B1 B1

		
6bii	$E_P = \frac{Qq}{4\pi\varepsilon_o D} = \frac{6.0\times10^{-9}\times3.0\times10^{-9}}{4\pi\times8.85\times10^{-12}\times0.60}$ $E_P = 2.7\times10^{-7}$ J	C1 A1
6biii	Any of the three below – B1) • Both <u>positively-charged/like charges</u> so <u>repel/force is repulsive</u> and <u>move away from each</u> other • <u>EPE of system is converted to KE/force same direction as velocity</u> so <u>speed increases</u> • Total momentum of system is conserved and zero, since the masses are equal so velocities always equal and opposite <u>Force decreases with distance</u>, so <u>acceleration decreases</u> / speed up with <u>decreasing rate</u>	B1 B1
7a	A photon is a <u>quantum</u> of <u>electromagnetic</u> radiation. The energy (of a photon) is dependent only on the frequency of radiation / Each photon has an energy given by the product of Planck constant and the frequency of the radiation / Each photon carries an energy = *hf*	B1 B1
7bi	Light is re-emitted in all directions / only part of the re-emitted light is in the direction of the (original) beam.	B1
7bii	Energy of photon released from the transition (largest energy difference so longest wavelength) $\frac{hc}{\lambda} = \frac{6.63\times10^{-34}\times3.00\times10^{8}}{658\times10^{-9}} = 3.0227\times10^{-19}\,\text{J}$ $E_3 - E_2 = 3.0227\times10^{-19}\,\text{J}$	C1

	$E_3 = 3.0227\times10^{-19} + \left(-3.40\times1.60\times10^{-19}\right) = -2.42\times10^{-19}$ J	A1
7biii	Electric potential energy is negative due to <u>negative work done</u> by an <u>external force</u> in <u>moving the electron from infinity to that point</u>. This is a result of the <u>attractive nature of the electric force</u> between the negatively charged electron and the positively charged nucleus.)	B1
8ai	work function = hc/λ_0 where λ_0 is the threshold wavelength of gallium arsenide $1.4 \times 1.60\times10^{-19} = 6.63\times10^{-34}\ (3.00\times10^{8}) / \lambda_0$ $\lambda_0 = 8.879\times10^{-7}$ = 890 nm	C1 A1
8aii	From Wien's law, $\lambda_{peak} = b/T$ $= 2.898\times10^{-3} / (273.15 + 37)$ $= 9.655\times10^{-6}$ m = 9655 nm Since the peak wavelength of radiation from a human body is <u>larger than</u> the threshold wavelength, (it has lower energy than work function) and **<u>no photoelectron</u>** can be emitted.	M1 A1
8b	The photoelectron was accelerated by an electric potential difference of 150 V. Gain in kinetic energy of photoelectron = work done by electric field on photoelectron $\frac{1}{2} m v^2 - 0 = e\Delta V$ $v = \sqrt{\frac{2e\Delta V}{m}}$ $= \sqrt{\frac{2(1.60\times10^{-19})150}{9.11\times10^{-31}}}$ $= 7.2587\times10^{6}$ $= 7.3\times10^{6}$ m s^{-1}	C1 A1
8ci	It ensures that every incoming electron strikes the channel wall upon entry / does not pass straight through without striking the wall	B1
8cii	No. of wall collisions, $n = \frac{L}{2d}\sin\theta$ $= \frac{480}{2(8.0)}\sin 13°$ $= 6.75 = 6$ Since 3 electrons are emitted at each collision site, electron gain = $(3)^6 = 729$	M1 A1
8d	1. The electrons need to <u>have high energy</u> in order to penetrate deep into the phosphor lattice. 2. High p.d. across a short distance ensures the electron <u>do not spread laterally</u> and illuminate adjacent spots on the screen.	B1 B1

8ei	minimum energy of each high-energy electron hits the phosphor screen = $e\Delta V$ = 4500 eV (assuming that the energy of the electron emerging from MCP is zero) Since each high-energy electron can create 450 electron-hole pairs, average energy to create one electron-hole pair = 4500 / 450 = 10 eV	C1 A1
8eii	total useful energy emitted as light = nhc/λ = 350 $(6.63\times10^{-34})(3.00\times10^{8}) / (545\times10^{-9})$ = 1.2773×10^{-16} minimum energy of incident electron = 4500 eV = 4500 (1.60×10^{-19}) = 7.20×10^{-16} maximum internal quantum efficiency = total useful energy / minimum input energy = $1.2773\times10^{-16} / 7.20\times10^{-16}$ = 0.1774 = 0.17	 C1 A1
8eiii	For two spots to be distinguishable, the angular separation has to be greater than the Rayleigh criterion. Rayleigh criterion, $\theta \approx \lambda / b$ = $545\times10^{-9} / 7.0\times10^{-3}$ = 7.79×10^{-5} rad Since the angular separation of two spots is <u>larger than the Rayleigh criterion</u> of the eye, they <u>can be distinguished</u> by the observer.	 M1 A1

8f

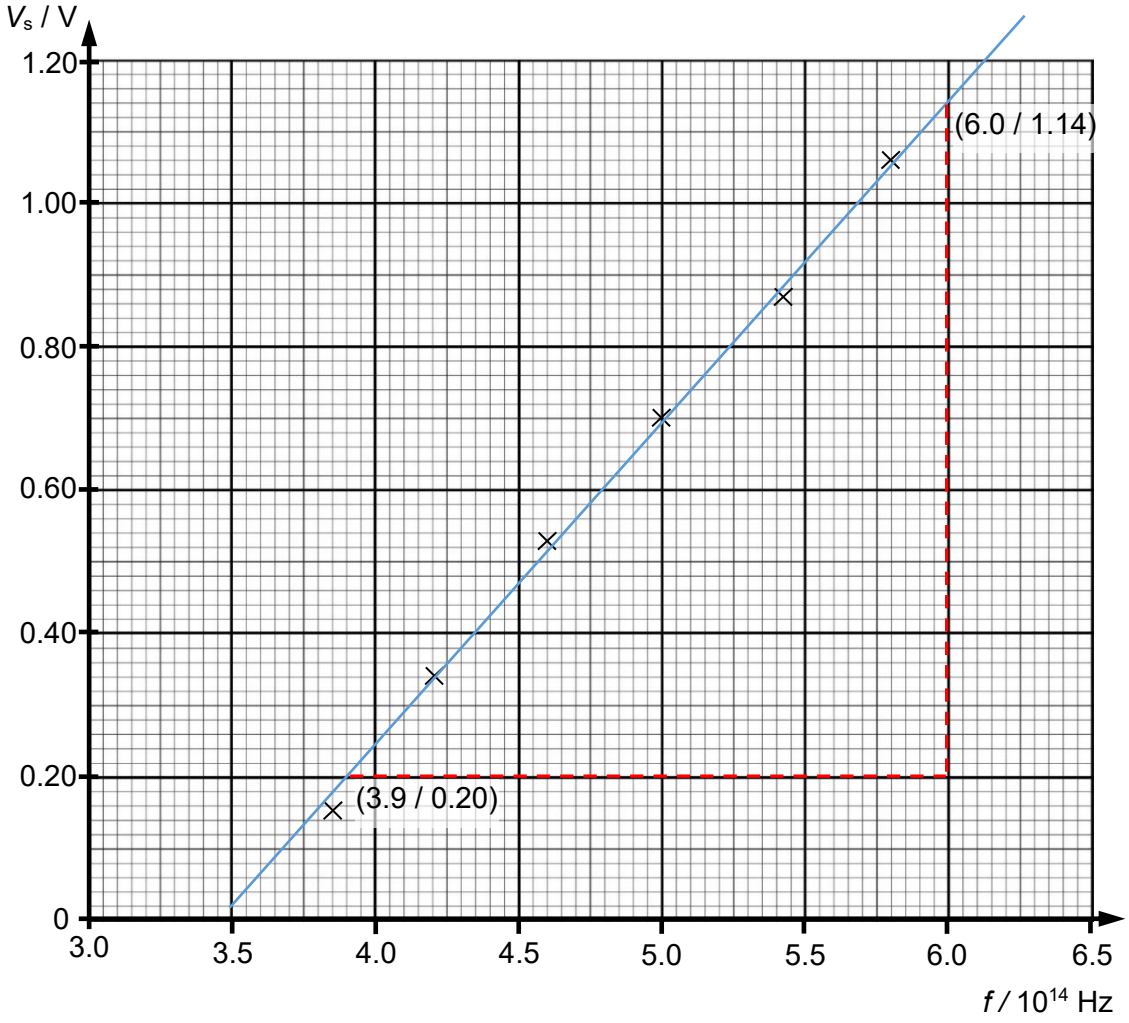


Figure 8.3

draw best-fit line with fair scatter of points above and below the whole line, as well as top and bottom half of the line. M1

gradient = $\frac{1.14 - 0.20}{(6.0\text{-}3.9) \times 10^{14}}$ M1

$= 4.476 \times 10^{-15}$

Rearranging the equation, we have

$$V_s = (h/e)f - \phi/e$$

gradient = h/e

$4.476 \times 10^{-15} = h / (1.60 \times 10^{-19})$

$h = 7.1616 \times 10^{-34}$

$= 7.2 \times 10^{-34}$ J s A1